DHARMSINH DESAI UNIVERSITY, NADIAD

FACULTY OF MANAGEMENT AND INFORMATION SCIENCE

B.C.A. SEMESTER VI

SUBJECT: (25BCA601)SYSTEM DEVELOPMENT PROJECT

Student Name	Himanshu D Ladha
Roll No	BCA045
Student ID	23BCUOS055

Project Definition: : RestoFlow is a full-stack Restaurant Management Platform built using .NET 10 Core and React JS, offering role-based access for Admins, Restaurant Owners, and Customers. It includes features like meal management, ordering, payments, and restaurant reviews to streamline restaurant operations and deliver a seamless digital dining experience.

Company Timing: 11:00 AM to 6:00 PM

Week	Date	Day	Name of the Topic	Signature
Second Week	8-12-2025	Monday	Installation, connection & working on the database	
	9-12-2025	Tuesday	Database design, Entity Framework, Framework Responsibility, configuration & API development	
	10-12-2025	Wednesday	Understanding framework components, functions and context	
	11-12-2025	Thursday	OOPS concept with examples	
	12-12-2025	Friday	Create table and register the user	

External Guide Name:

Signature

Internal Guide Name: Prof. Rupal Patel

Signature

Date :- 8-12-2025

- Installation of SQL Express
- Connection with SQL Express to SSMS
- Database Creation
- Table Creation
- Learning of DDL, DML and DQL SQL Languages.
- Types of SQL Joins.
- Learning of Data first and Code first approaches for Code to Database integration.

Date :- 9-12-2025

- Continued database table creation for the RMS, focusing on user-related tables:
    - User, UserLogin, UserType, Restaurant, Meal, PaymentMode, Other supporting tables.
- Inserted dummy values for testing purposes.
- Integrated Entity Framework Core by creating a separate project: RMS.EntityFramework (Database-First Approach).
- Installed necessary NuGet packages and dotnet-ef tool for EF operations.
- Configured connection string in appsettings.json.
- Used Reverse Engineer Scaffold to generate Entity Framework models and DbContext from the existing database.
- Created API controllers to fetch and manipulate user details.
- Learned about Entity Framework's role in:
    - Connecting to SQL Server
    - Mapping database tables to C# classes
    - Managing relationships and foreign key constraints
    - Tracking data changes
    - Running LINQ queries to interact with the database

Date :- 10-12-2025

- Learn the meanings of the entity framework components.
- Key Concepts and Terminology
- DbContext and Its Role, DbSet and Table Mapping
- Model Classes and Relationships
- Change Tracking and State Management
- Each functions and related context things.
- And watched the Youtube videos on the JavaScript.

Date :- 11-12-2025

- Learn the concepts of Object Oriented Programming.
- Encapsulation, Abstraction, Inheritance, Polymorphism.
- Implemented this in the project.

Date :- 12-12-2025

- Created the new tables required for the project.
- Added the some of the api’s in the project , to test the backend side.
- Register the user using the swagger, from the backend